Namioka–Phelps spaces satisfy the nonlinear dual fixed-point theorem

Candidate full solution to Question 1 of arXiv:1903.12123

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Status. Candidate full affirmative answer, likely valid; human review recommended. The result answers Question 1 only.

1 Source question

Andrzej Wiśnicki, *Around the nonlinear Ryll–Nardzewski theorem*, arXiv:1903.12123v2, asks on page 10:

Question 1. Is it true that Corollary 3.2 remains true for Namioka–Phelps spaces?

Here Corollary 3.2 says that if X is Asplund, $Q \subset X^*$ is weak-star compact and convex, and a dynamical system on Q is nonexpansive and norm-distal, then it has a fixed point [1]. The locally convex conventions come from Glasner–Megrelishvili [2]: the standard uniformity ξ^* on V^* is uniform convergence on bounded subsets of V, and a Namioka–Phelps space is one for which every equicontinuous subset of V^* is (weak*, ξ^*)-fragmented.

2 Definitions and statement

Let V be a Hausdorff locally convex space. For every bounded $B \subset V$, put

$$p_B(x^*) = \sup_{v \in B} |x^*(v)| \qquad (x^* \in V^*).$$

These seminorms generate ξ^*. A semigroup action (S, Q) is ξ^*-nonexpansive if

$$p_B(sx - sy) \le p_B(x - y) \quad (s \in S,\ x, y \in Q)$$

for every bounded $B \subset V$. It is ξ^*-distal if for every $x \neq y$ there are a bounded $B \subset V$ and $\varepsilon > 0$ such that $p_B(sx - sy) > \varepsilon$ for all $s \in S$.

Theorem 1 (Affirmative answer to Question 1)**.** *Let V be a Namioka–Phelps locally convex space and let $Q \subset V^*$ be a nonempty equicontinuous weak-star compact convex set. Suppose S acts on Q by weak-star continuous maps and the action is ξ^*-nonexpansive and ξ^*-distal. Then S has a common fixed point in Q.*

If V is barrelled, every weak-star compact subset of V^* is equicontinuous, so the explicit equicontinuity clause may then be omitted.

single point x and $sx = x$ for every $s \in S$. $\square$

There is a natural generalization of Asplund Banach spaces, introduced in [20, Definition 4.1] (see also [12, Definition 1.10]): a locally convex space (X, τ) is called a Namioka–Phelps space if every equicontinuous subset K in X^* is (weak*, ξ^*)-fragmented, where ξ^* denotes the natural uniform structure of X^*. It is shown in [20] that the class of Namioka–Phelps spaces contains in particular Fréchet differentiable spaces, semireflexive spaces and nuclear spaces, and is closed under taking subspaces, products and direct sums (see also [12, Remark 1.12]). Since the weak compactness of Q is applied in the proof of Theorem 4.1 in a substantial way to show the τ-separability of K, it is not clear how to extend Corollary 3.2 to the case of locally convex spaces. This leads to the following natural questions.

Question 1. Is it true that Corollary 3.2 remains true for Namioka–Phelps spaces?

More generally, we can ask:

Question 2. Do there exist nonlinear counterparts of Theorems 1.5 and 1.6 in [12]?

Figure 1: The two source questions on page 10. This packet answers Question 1.

3 Proof intuition

The Banach-space proof in [1] uses the Radon–Nikodým property first to fragment a minimal subsystem and later to make that subsystem norm compact. Namioka–Phelps fragmentation supplies the first step directly. For the second step, one does not need a countable dense set: fix one bounded $B \subset V$ and one scale. Fragmentation produces a positive-measure set of small p_B-diameter. Nonexpansivity moves such sets to neighborhoods of all orbit points without decreasing their invariant measure, so only finitely many orbit points can be separated at that scale. Repeating this independently for all B proves ξ^*-total boundedness. A short net argument supplies the completeness that the source identified as problematic.

4 Proof

We isolate the completeness observation.

Lemma 1. *Every equicontinuous weak-star compact subset $K \subset V^*$ is complete for the uniformity induced by ξ^*.*

Proof. Let (x_i) be a ξ^*-Cauchy net in K. Passing to a subnet, weak-star compactness gives $x_{i_\alpha} \to x \in K$ weak-star. Fix a bounded $B \subset V$ and $\varepsilon > 0$. Eventually $p_B(x_i - x_j) \le \varepsilon$ for all sufficiently large i, j. Fix such an i and let $j = i_\alpha$ tend weak-star to x. For every $v \in B$,

$$|(x_i - x)(v)| = \lim_\alpha |(x_i - x_{i_\alpha})(v)| \le \varepsilon.$$

Taking the supremum over $v \in B$ gives $p_B(x_i - x) \le \varepsilon$. Thus $x_i \to x$ in ξ^*. $\square$

Proof of Theorem 1. By Zorn's lemma, replace Q by a minimal nonempty weak-star compact convex S-invariant subset. Inside it choose a minimal nonempty weak-star compact S-invariant subset K, not assumed convex.

1. Lifting distality. We claim that (S, K) is weak-star distal. Otherwise there are distinct $x, y \in K$, a net $(s_\alpha) \subset S$, and $u \in K$ such that $s_\alpha x$ and $s_\alpha y$ both converge weak-star to u. By ξ^*-distality, choose an entourage E of ξ^* such that $(sx, sy) \notin E$ for all $s \in S$. Since V is Namioka–Phelps and K is equicontinuous, fragmentation gives a weak-star open U meeting K whose intersection with K is E-small (after replacing E by a smaller symmetric entourage if necessary). Minimality gives $K = \overline{Su}^{\text{weak}^*}$, so some $s_0 u$ lies in $U \cap K$. Both $s_0 s_\alpha x$ and $s_0 s_\alpha y$ eventually lie in $U \cap K$, contrary to the choice of E.

The standard Furstenberg–Namioka invariant-measure theorem for compact distal systems now gives an S-invariant Radon probability measure μ on K [1, 2]. Replacing K by $\operatorname{supp} \mu$ and using minimality, we have

$$\operatorname{supp} \mu = K, \qquad s(K) = K \quad (s \in S).$$

Indeed invariance shows first $s(K) \subset K$ and then that the weak-star compact set $s(K)$ has full measure, forcing the reverse inclusion.

2. Total boundedness from packing. Fix a bounded $B \subset V$ and $\varepsilon > 0$. Fragmentation supplies a weak-star open U meeting K such that

$$\operatorname{diam}_{p_B}(U \cap K) < \varepsilon/2.$$

Choose $x \in U \cap K$. Because $x \in \operatorname{supp} \mu$, the set

$$A = \{y \in K : p_B(x - y) < \varepsilon/2\}$$

has measure $\delta > 0$. Nonexpansivity and invariance imply, for every $s \in S$,

$$\mu\{y \in K : p_B(sx - y) < \varepsilon/2\} \geq \delta.$$

The displayed neighborhoods belonging to orbit points that are pairwise p_B-separated by at least ε are disjoint. Hence there can be at most $1/\delta$ such orbit points. A greedy maximal separated family is therefore finite and is an ε-net for Sx. As B and ε were arbitrary, Sx is ξ^*-totally bounded.

By Lemma 1, the ξ^*-closure C of Sx in K is ξ^*-compact. The identity from ξ^* to weak-star is continuous, so C is weak-star compact and therefore weak-star closed. Minimality gives $K = \overline{Sx}^{\text{weak}^*} \subset C$, while $C \subset K$ is immediate. Thus K is ξ^*-compact.

3. Normal-structure contradiction. Suppose K has more than one point. Some bounded $B \subset V$ satisfies

$$r = \sup_{x,y \in K} p_B(x - y) > 0.$$

The locally convex DeMarr lemma (the seminorm version of the usual compact-set normal-structure lemma) gives $u \in \overline{\operatorname{co} K}^{\xi^*}$ and $r_0 < r$ with

$$\sup_{y \in K} p_B(u - y) \leq r_0.$$

Since Q is convex and weak-star closed, $u \in Q$. Define

$$Q_0 = \{z \in Q : p_B(z - y) \leq r_0 \text{ for every } y \in K\}.$$

The seminorm p_B is weak-star lower semicontinuous (it is a supremum of weak-star continuous functions), so Q_0 is weak-star compact and convex. It is nonempty, contains u, and is proper because $r_0 < \operatorname{diam}_{p_B} K$. Finally, $s(K) = K$ and nonexpansivity give $s(Q_0) \subset Q_0$ for every $s \in S$. This contradicts the minimality of Q.

Therefore K is a singleton, and its unique point is fixed by every element of S. $\square$

5 Verification, scope, and novelty

The proof avoids the source's separability bottleneck: total boundedness is proved one strong-dual seminorm at a time, and completeness is supplied by Lemma 1. No metrizability, countable bounded-set base, or completeness of the whole strong dual is assumed.

Question 2 of the source is broader and is not claimed here. The result also retains equicontinuity of Q, as required in the standard locally convex dual fixed-point formulation of [2].

A bounded novelty check searched the run indexes, arXiv, and the web using the exact question, source title and id, and combinations of "Namioka–Phelps", "nonlinear", "nonexpansive", "distal", and "fixed point". Later local arXiv sources citing the paper were also checked. No later exact answer was found. This is not exhaustive bibliographic certification.

Human review recommendation

Check especially (i) Lemma 1, including its use of nets, and (ii) the packing argument for a nonmetrizable strong-dual uniformity. These are the two steps replacing the weak-compactness/separability mechanism in the source.

References

[1] A. Wiśnicki, *Around the nonlinear Ryll–Nardzewski theorem*, Math. Ann. 377 (2020), 267–279; arXiv:1903.12123.

[2] E. Glasner and M. Megrelishvili, *On fixed point theorems and nonsensitivity*, Israel J. Math. 190 (2012), 289–305; arXiv:1007.5303.